

Dietary carbohydrates are used to synthesize liver glycogen, which exists as particles of various sizes. During the synthesis of a new glycogen particle, a glucose molecule is linked to the protein glycogenin, after which additional glucose molecules are linked to the first glucose and to each other to form regions that are successively farther from glycogenin. Glycogen synthesis is catalyzed by the enzyme glycogen synthase, whose activity is inhibited by phosphorylation of three carboxy-terminal serine residues. During the postabsorptive state (ie, after carbohydrate absorption from a meal is finished), hormone-stimulated release of glucose residues from liver glycogen stores help maintain adequate glucose levels in the blood.

Although the important role of liver glycogen in whole-body glucose metabolism has long been appreciated, much remains unknown about the regulation of liver glycogen synthesis and breakdown. Based on previous studies suggesting a relationship between glycogen particle size and the rate of glycogenolysis, a team of researchers measured the average sizes of liver α and β glycogen particles in wild-type (WT) mice (Table 1).

Table 1 Dimensions of Liver Glycogen Particles

	β particles	α particles
Diameter (nm)	20	200
Surface area (nm ²)	1.26×10^3	1.26×10^5
Volume (nm ³)	4.19×10^3	4.19×10^6

To better understand the role of the protein glycogenin in glycogen synthesis, the researchers also mated male and female mice that were heterozygous for glycogenin expression (GN +/–) to produce heterozygous offspring as well as mice with WT and knockout (GN –/–) genotypes. Relative glycogenin gene expression (ie, mRNA levels) was measured in liver and heart samples from the mice (Table 2). Glycogenin was undetectable in western blots of tissue samples from knockout mice.

Table 2 Gene Expression (mRNA Levels Relative to WT) in Liver and Heart Samples.

	WT	GN (+/–)	GN (–/–)
Liver	100	25	4
Heart	100	40	5

The researchers also measured glycogen concentrations in liver and heart samples from the mice. The results are presented in Figure 1.

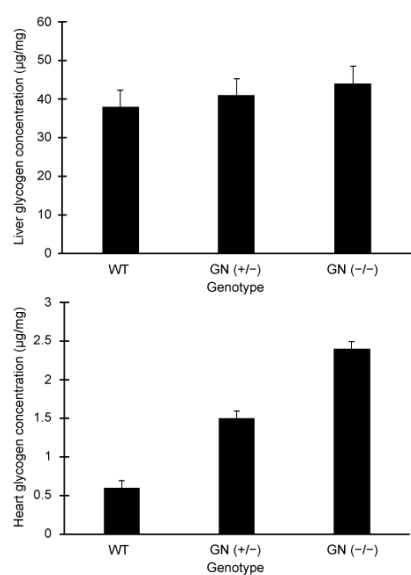
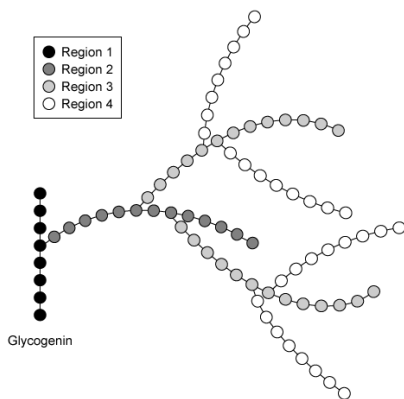


Figure 1 Liver and heart glycogen concentrations

For the portion of a glycogen β particle shown below, which of the regions contain the smallest ratio of $\alpha(1\rightarrow6)$ glycosidic bonds to $\alpha(1\rightarrow4)$ glycosidic bonds?

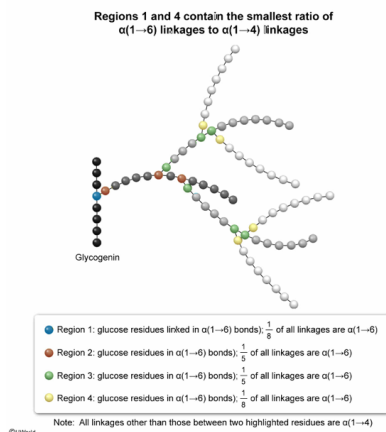


- ☐ A. Regions 1 and 2 only [8%]
- ✓ ☒ B. Regions 1 and 4 only [41%]
- ☐ C. Regions 2 and 3 only [11%]
- ☐ D. Region 1 only [38%]

Correct

42% answered correctly

Explanation:



Glycogen is a branching carbohydrate polymer composed of glucose monomers and serves as a reservoir of glucose that can be drawn upon when needed. The anomeric carbon (ie, carbon 1) of glucose can be linked to other atoms in one of two configurations, α or β . The glucose monomers in glycogen are α anomers, linked to one another by glycosidic bonds of two types, $\alpha(1\rightarrow4)$ and $\alpha(1\rightarrow6)$. The $\alpha(1\rightarrow4)$ linkages occur in linear segments whereas the $\alpha(1\rightarrow6)$ linkages occur at branch points.

The question presents a portion of a glycogen molecule and asks for the regions with the lowest ratio of $\alpha(1\rightarrow6)$ to $\alpha(1\rightarrow4)$ bonds. One eighth of the linkages between glucose monomers in both **region 1** and **region 4** are $\alpha(1\rightarrow6)$ linkages. A greater fraction (ie, one fifth) of the linkages in regions 2 and 3 are $\alpha(1\rightarrow6)$ glycosidic bonds. Therefore, the $\alpha(1\rightarrow6)$ to $\alpha(1\rightarrow4)$ ratio is **smallest for glucose linkages in regions 1 and 4**.

(Choices A and C) One fifth (ie, 3 of 15 linkages in region 2 and 6 of 30 linkages in region 3) of the total linkages in regions 2 and 3 are via $\alpha(1\rightarrow6)$ glycosidic bonds, which is greater than the one eighth ratio in regions 1 and 4.

(Choice D) One out of eight linkages in region 1 and region 4 are linked to glucose residues in other regions via $\alpha(1\rightarrow6)$ glycosidic bonds.

Educational objective:

Branching glucose residues in glycogen are linked via $\alpha(1\rightarrow6)$ glycosidic bonds; the other glucose residues are linked by $\alpha(1\rightarrow4)$ glycosidic bonds.